MATH **152** **QUIZ 11** NAME Solutions

Mrs. Bonny Tighe 25 points

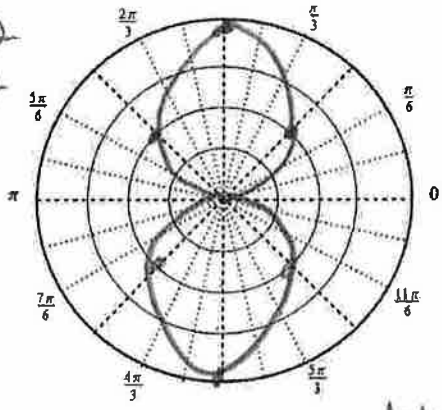
10.4 TA _____ Monday 5/13/2024

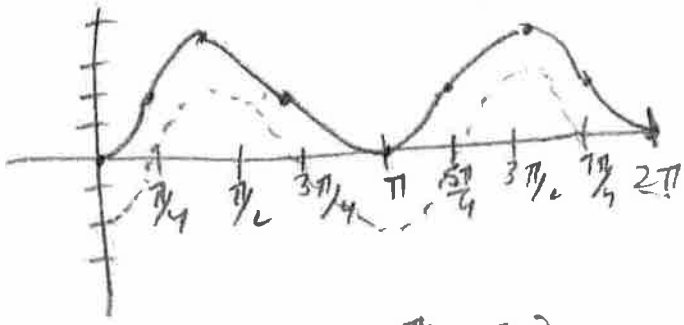
1. Find the slope of the tangent line to the curve $r = 2 - 2\cos 2\theta$ when $\theta = \pi/4$

GRAPH

$x = r\cos\theta$

$y = r\sin\theta$





$r(\pi/4) = 2 - 2\cos\pi/2 = 2$

$r'(\theta) = 4\sin 2\theta \quad r'(\pi/4) = 4\sin\pi/2 = 4$

$$dy/dx = \frac{r\cos\theta + \sin\theta\, dr/d\theta}{r(-\sin\theta) + \cos\theta\, dr/d\theta}$$

$$\frac{2(\sqrt{2}/2) + \sqrt{2}/2\,(4)}{2(-\sqrt{2}/2) + \sqrt{2}/2\,(4)} = \frac{\sqrt{2} + 2\sqrt{2}}{-\sqrt{2} + 2\sqrt{2}} = \frac{3\sqrt{2}}{+\sqrt{2}} = 3$$

2. Find the length of the polar curve $r = \sin^4\left(\frac{\theta}{4}\right)$ $\quad L = \int_a^b \sqrt{r^2 + \left(dr/d\theta\right)^2}\, d\theta$

$dr/d\theta = 4\sin^3\left(\frac{\theta}{4}\right)\cdot\frac{1}{4}\cos\left(\frac{\theta}{4}\right)$ $\quad L = \int_0^{2\pi} \sqrt{\sin^8\left(\frac{\theta}{4}\right) + \sin^6\frac{\theta}{4}\cos^2\frac{\theta}{4}}\, d\theta$

$$L = \int_0^{2\pi} \sqrt{\sin^6\left(\frac{\theta}{4}\right)\left(\sin^2\frac{\theta}{4} + \cos^2\theta/4\right)}\, d\theta = \int_0^{2\pi} \sin^3\left(\frac{\theta}{4}\right) d\theta$$

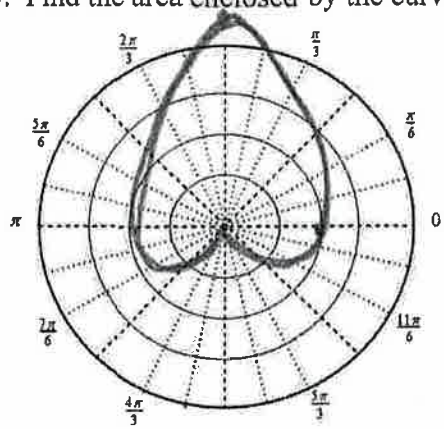
$$= \int \left(1 - \cos^2\frac{\theta}{4}\right)\left(-\sin\frac{\theta}{4}\right) d\theta = \int \left(\cos^2\frac{\theta}{4} - 1\right)\left(-\sin\frac{\theta}{4}\right) d\theta$$

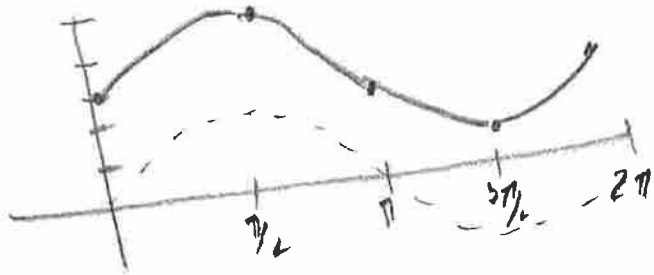
$$\frac{4}{3}\cos^3\left(\frac{\theta}{4}\right) - 4\cos\frac{\theta}{4}\Big]_0^{2\pi} = \left(\frac{4}{3}\cos^3\left(\frac{\pi}{2}\right) - 4\cos\left(\frac{\pi}{2}\right)\right) - \left(\frac{4}{3}\cos^3 0 - 4\cos 0\right)$$

$$-\left(\frac{4}{3} - 4\right)$$

$8/3$

3. Find the area enclosed by the curve $r = 3 + 2\sin\theta$. **Sketch the graph.**





$$\int_0^{2\pi} \frac{1}{2}(3+2\sin\theta)^2\,d\theta = \frac{1}{2}\int_0^{2\pi} 9 + 12\sin\theta + 4\underbrace{\sin^2\theta}_{(\frac{1}{2}-\frac{1}{2}\cos 2\theta)}\,d\theta$$

$$\frac{1}{2}\int_0^{2\pi}(9+12\sin\theta + 2 - 2\cos 2\theta)\,d\theta = \frac{1}{2}\left[9\theta - 12\cos\theta + 2\theta - \sin 2\theta\right]_0^{2\pi}$$

$$\frac{1}{2}\left[(11(2\pi) - 12\cos 2\pi - \sin 4\pi) - (0 - 12 + 0 - 0)\right] = \frac{1}{2}(22\pi)$$

$$11\pi$$

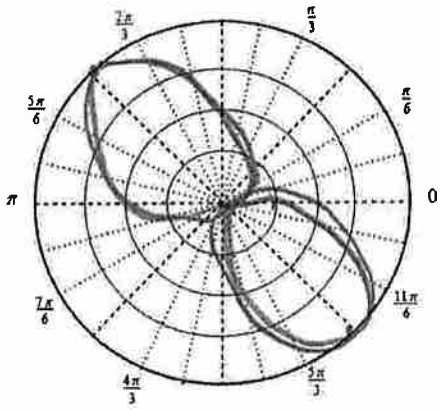
4. Find the length of the polar curve $r = e^{4\theta}$ on the interval $0 \le \theta \le 2\pi$

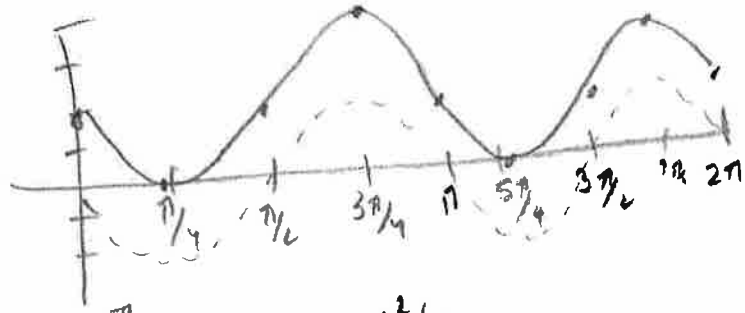
$$L = \int_a^b \sqrt{r^2 + \left(dr/d\theta\right)^2}\,d\theta \qquad dr/d\theta = e^{4\theta}(4) =$$

$$L = \int_0^{2\pi} \sqrt{e^{8\theta} + 16e^{2\theta}}\,d\theta = \int_0^{2\pi} \sqrt{17}\,e^{4\theta}\,d\theta$$

$$\sqrt{17}\,\frac{e^{4\theta}}{4}\Big]_0^{2\pi} = \boxed{\sqrt{17}\left(\frac{e^{8\pi}}{4} - \frac{1}{4}\right)}$$

5. Sketch the graph $r = 2 - 2\sin 2\theta$ and find the enclosed area.





$$2\int_0^{\pi} \tfrac{1}{2}(2-2\sin 2\theta)^2\,d\theta$$

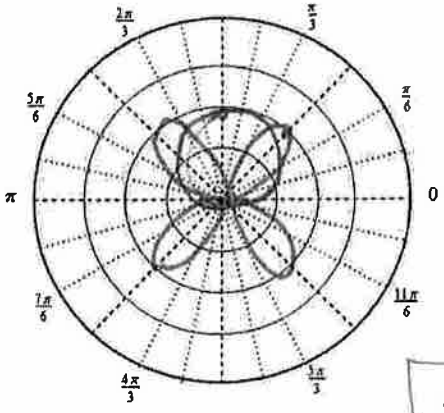
$$\int 4 - 8\sin 2\theta + 4\sin^2 2\theta \, d\theta = \int (4 - 8\sin 2\theta + 2 - 2\cos 4\theta)\, d\theta$$

$$6\theta + 4\cos 2\theta - \tfrac{1}{2}\sin 4\theta \Big]_0^{\pi}$$

$$(6\pi + 4\cos 2\pi - \tfrac{1}{2}\sin 4\pi) - (0 + 4\cos 0 - 0)$$

6π

6. Find all points of intersection for the curves $r = 2\sin 2\theta$ *and* $r = 2\sin\theta$ GRAPH



$$\cancel{2}(2\sin\theta\cos\theta) = \cancel{2}\sin\theta$$

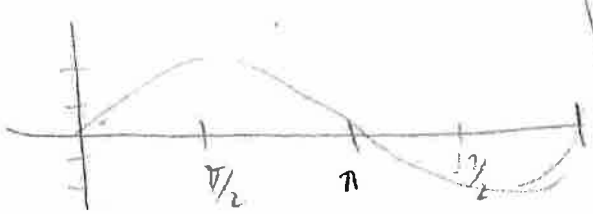
$$2\sin\theta\cos\theta - \sin\theta = 0$$

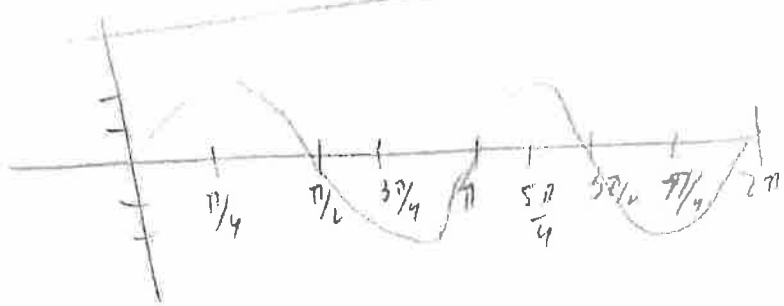
$$\sin\theta(2\cos\theta - 1) = 0$$

$\sin\theta = 0$ $\quad$ $\cos\theta = 1/2$

$\theta = 2\pi n$ $\quad$ $\theta = \pi/3 + 2\pi n$ $\quad$ $\frac{5\pi}{3} + 2\pi n$

$$\boxed{(0, 2\pi n) \quad (1, \pi/3 + 2\pi n) \quad (-1, 5\pi/3 + 2\pi n)}$$



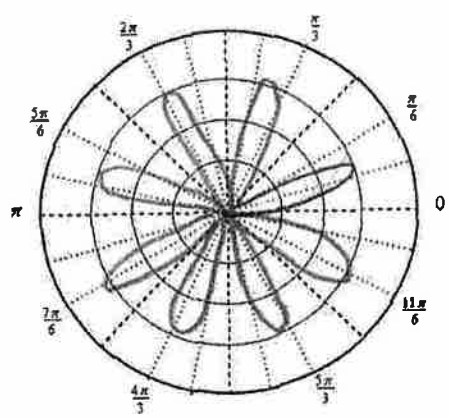


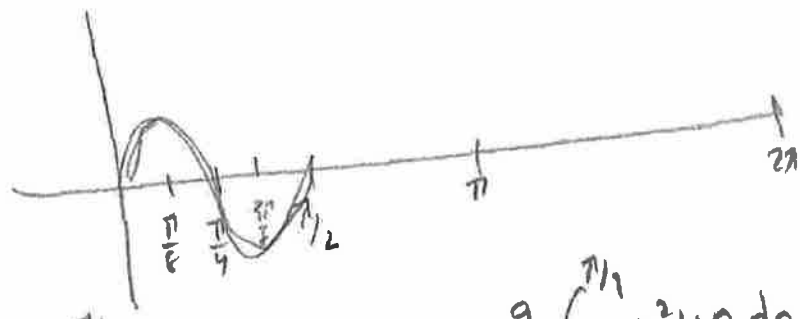
7. Find the area of the region that is bounded by the given curve and lies in the specified sector. $r = e^{-\theta/3}$ $\pi/2 \le \theta \le \pi$

$$\int_{\pi/2}^{\pi} \frac{1}{2}\left(e^{-\theta/3}\right) d\theta = \frac{1}{2}(-3)e^{-\theta/3}\Big]_{\pi/2}^{\pi}$$

$$-\frac{3}{2}\left[e^{-\pi/3} - e^{-\pi/6}\right]$$

9. Graph and find the area of the region enclosed by one loop of the curve $r = 3\sin 4\theta$





$$\int_0^{\pi/4} \frac{1}{2}(3\sin 4\theta)^2 d\theta = \frac{9}{2}\int_0^{\pi/4} \sin^2 4\theta \, d\theta$$

$$\frac{9}{2}\int\left(\frac{1}{2} - \frac{1}{2}\cos 8\theta\right) d\theta = \frac{9}{2}\left[\frac{1}{2}\theta - \frac{1}{16}\sin 8\theta\right]_0^{\pi/4}$$

$$\frac{9}{2}\left[\left(\frac{1}{2}\left(\frac{\pi}{4}\right) - \frac{1}{16}\sin 2\pi\right) - (0)\right] = \frac{9}{2}\left[\frac{\pi}{8}\right] = \frac{9\pi}{16}$$